AI Powered Document Verification - A Behavioral and Multi Domain Approach to Automated Fraud Detection

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Abstract*— To combat identity fraud and misuse of stolen documents To combat identity fraud, phishing, and misuse of stolen or forged documents. This paper introduces a unified framework for secure and accurate document verification by integrating a behavioral authentication layer with an AI-powered forgery detection pipeline. The system begins with a Human-AI Interfacing Mechanism, the UserKey Protocol, which uses contextual questions to ensure only legitimate users initiate the verification process. Subsequently, documents are securely retrieved via DigiLocker APIs and undergo a rigorous analysis. The core of the system is an AI pipeline that preprocesses documents, extracts multi-domain features including visual, textual, structural, and biometric indicators, and classifies them as either authentic or forged. Leveraging a hybrid model powered by Convolutional Neural Networks (CNNs), Optical Character Recognition (OCR), and advanced preprocessing for primary classification and Support Vector Machines (SVMs) and Random Forests for specialized feature analysis, the system achieves a high classification accuracy of 98%. The system's robustness is enhanced through data augmentation and a feedback loop for continuous model retraining. This end-to-end solution provides real-time, scalable, and automated fraud detection for high-stakes applications in government, finance, and other enterprise environments. Future work will explore integration with blockchain for tamper-proof auditing and enhancement with attention-based architectures to improve robustness against subtle forgeries aligning with global standards for data security and compliance.

Keywords— Document Verification, Deep Learning, Fraud Detection, Image Processing, CNN,

I. Introduction

Fraudsters use various sophisticated methods to manipulate documents and bypass verification systems. These methods are categorized based on the nature of the deception. Completely fake documents are created from scratch, lacking the genuine security features of an authentic ID. In contrast, false documents are genuine IDs that have been stolen and are used by an impersonator. A more prevalent type is a modified document, which involves altering an original document. This includes a variety of techniques such as Card Edit Checks, where textual data like names or dates of birth are changed, and Face Morphing and Face Editing, where the ID photo is manipulated to resemble the fraudster. Other forms of this fraud include Signature Forgery, Document Image Fraud—altering scanned documents with software—and Metadata Tampering, which involves changing a file's digital footprint. Additionally, with the advent of generative AI, a new threat has emerged: Generated Document Fraud, where AI tools are used to create entirely new, convincing but fraudulent documents

TABLE I. DOCUMENT FRAUD TYPE

Fraud Type	Description	Key Characteristics	Detection Method
Completely Fake (Counterfeit)	Documents created entirely from scratch.	Lacks genuine security features (e.g., QR codes, holograms).	AI analysis of document format, landmarks, and security features.
False (Stolen)	Genuine, authentic documents that have been stolen.	Document is real, but the user is an impersonator.	Facial recognition and liveness detection are required.
Modified (Forged/Altered)	Original documents that have been altered to deceive verification.	Subtle changes to text, images, or security features.	AI-powered checks for altered text, images, signatures, and document metadata.

Document verification has traditionally relied on manual inspections, which are labor-intensive and error-prone [1], [2]. OCR-based methods [3] improved automation but struggled with structural forgeries. With the advancement of CNN-based deep learning [4], [5], detection accuracy and scalability improved significantly. Blockchain integration [7], [8], [14] and emerging technologies such as GANs [9], [21], IoT [10], and adaptive AI [12] further enhance resilience against evolving threats. Current research emphasizes hybrid AI-blockchain frameworks and explainable AI to balance robustness, security, and interpretability. The proposed system integrates behavioral authentication, DigiLocker based secure document retrieval, and an AI-driven forgery detection pipeline. The workflow ensures that only authenticated users can access government-issued documents, which are then analyzed using advanced machine learning and deep learning algorithms to detect authenticity or forgery. In the digital era, document fraud has become a significant challenge across critical sectors like finance, healthcare, and government. The authenticity of documents such as passports, identity cards, and academic certificates is vital for maintaining trust and security. However, with the rise of sophisticated forgery techniques,

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traditional manual verification methods are increasingly unreliable, time-consuming, and unable to scale for high-volume tasks.Artificial Intelligence (AI), particularly deep learning, offers a powerful solution for automating and improving document verification. By leveraging Convolutional Neural Networks (CNNs) and advanced image processing, AI systems can identify subtle inconsistencies in fonts, signatures, watermarks, and security features that are often missed during manual inspection. This paper introduces a robust, AI-based document verification system designed to efficiently and accurately detect forged documents. The system combines a behavioral authentication layer with a comprehensive document analysis pipeline, delivering real-time verification with high accuracy.

UserKey is a contextual, dynamic, preference-based authentication mechanism that verifies the user based on personal behavioral patterns and responses to real-time contextual prompts. It acts as a soft biometric signature, unique to the user today's digital era, document fraud has emerged as a significant challenge impacting critical sectors such as finance, healthcare, and government. The authenticity of documents like passports, identity cards, academic certificates, and financial statements is vital for maintaining trust and security in various processes. However, with the advancement of forgery techniques, traditional manual verification methods have become increasingly unreliable, time-consuming, and unable to scale for high-volume verification tasks. Artificial Intelligence (AI), especially deep learning, provides promising solutions for automating and improving document verification. By utilizing Convolutional Neural Networks (CNNs) and advanced image processing techniques, AI-powered systems can identify subtle inconsistencies in fonts, signatures, watermarks, and embedded security features that often go unnoticed during manual inspection. This paper introduces a robust AI-based document verification system designed to efficiently and accurately detect forged documents. The system combines preprocessing pipelines, feature extraction, and classification models to deliver real-time verification with high accuracy. Additionally, future expansions include blockchain integration for tamper- proof record-keeping and support for mobile platforms to enable widespread accessibility. The following sections detail the system architecture, methodology, experimental results, and potential avenues for future enhancement.

II. Literature Review

The detection of document forgery and fraudulent verification processes has been widely studied, combining image processing, machine learning, deep learning, and emerging technologies such as blockchain and IoT.

Image Processing and Machine Learning Approaches: Chawla et al. [1] proposed a hybrid approach integrating image processing with machine learning to detect forgery, demonstrating that features such as texture and edge consistency can improve accuracy when combined with classifiers. Similarly, Cekmez et al. [5] developed an AI-based verification framework for secure online transactions. Comparative evaluations by Cao et al. [6] and Alzamil [13] indicated that traditional classifiers like Random Forests and SVMs are effective but lack the robustness of deep learning approaches.

Optical Character Recognition (OCR) and Text-Based Methods:OCR has been extensively used for textual verification in digital documents. Carter [3] and Patel and Mehta [19] showed how OCR enables the detection of tampering by analyzing font style, alignment, and spacing, although its effectiveness decreases in noisy or low-resolution inputs.

Deep Learning and Convolutional Models **:**CNN-based approaches have transformed forgery detection. Jia and Kong [4] pioneered CNN use in intrusion detection, inspiring adaptation for document analysis. Khan and Ali [18] applied convolutional auto encoders for deep feature extraction, while recent CNN frameworks consistently achieve high performance in identifying anomalies such as forged signatures, stamps, and watermarks.

Explainable AI (XAI) **:**Trust and interpretability have become crucial in fraud detection. Lewis [2] introduced explainable AI into document verification, while Smith and Doe [17] and Verma and Singh [20] proposed interpretable frameworks that enhance model transparency and end-user confidence in decision-making.

Blockchain-Based Verification:Blockchain integration provides immutable and tamper-proof verification. Patel [7] and Zhao [8] demonstrated blockchain's utility in authentication systems. Kumar and Singh [14] advanced decentralized verification frameworks, while Ahmed and Khan [26] highlighted its synergy with AI in cybersecurity.

Emerging Technologies: IoT, GANs, and Adaptive AI: Recent studies address advanced forgery and phishing attacks. Wang [9] and Chen and Wang [21] leveraged GAN-based models for detecting deepfake forgeries. Verma [10] proposed IoT-enabled AI for distributed fraud detection. Zhang [12] introduced adaptive AI models to ensure real-time and evolving fraud resistance in verification systems.

Applications and Domain-Specific Studies: AI-powered verification has been applied across multiple domains. Sharma and Gupta [16] applied AI for online KYC verification. Johnson and Lee [24] proposed self-sovereign identity frameworks, aligning with secure e-governance. Taylor and Brown [22] emphasized explainable AI for real-time fraud detection, while Singh and Kaur [23] demonstrated AI's role in practical fraud prevention. Garcia and Martinez [25] consolidated these trends in a comprehensive review.

I. A Survey of Existing Public Identity Document Datasets

Several publicly available datasets have been introduced for tasks such as recognition, classification, and restoration of identity documents captured by mobile devices. The SmartDoc dataset provides 10 training document samples, each recorded as a video clip along with a corresponding ground-truth image, and 37 test samples for document restoration benchmarking [26]. The LRDE dataset contains 100 videos of passports and visas under varied conditions and using different smartphones [27].

The MIDV family of datasets has become a standard benchmark. MIDV-500 contains 500 video clips across 50 document types, including ID cards, passports, and driving licenses, with five different backgrounds and two mobile devices. It supports tasks such as OCR, face detection, and document-type classification [28]. MIDV-2019 extended this

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dataset with distorted and low-light captures for each document type [29]. MIDV-2020 further expanded the benchmark to 1000 unique samples spanning 10 document categories, significantly increasing its applicability [30].

Datasets focusing on fraudulent identity documents are more recent. The FMIDV dataset introduced seven forged versions for each MIDV-2020 sample using guilloche copy-and-move attacks, though it is limited to this single fraud type [31]. SIDTD extended the scope by generating 1222 forged documents using crop-and-move and inpainting techniques, thereby modeling additional attack scenarios [32].

Other regional datasets have also been proposed. For instance, the Brazilian Identity Document (BID) dataset consists of 28,800 images across eight document types, created by altering authentic documents. However, these alterations (e.g., blurred portraits) reduce its effectiveness for fraud detection involving facial images [33].

To address scalability and fraud coverage, the IDNet benchmark was recently introduced. IDNet is entirely synthetic and privacy-preserving, comprising 837,060 documents across 20 types. Each authentic sample is paired with six fraudulent variants, covering face morphing, portrait substitution, text alteration, crop-and-move, and inpainting-based frauds. Metadata is provided in JSON files, describing fields such as document identifiers, personal attributes, fraud type, and parameters. IDNet represents the largest publicly available benchmark for document fraud detection to date [34]

A. Dataset Preparation

To evaluate the proposed Document Verification System, a dataset comprising 1,500 images of official documents was assembled. These documents were obtained from publicly available sources or synthetically generated to reflect realistic forgery scenarios. The dataset includes Aadhaar cards, PAN cards, driver's licenses, and academic certificates, and experience certificate bank card.

Forgery simulations were introduced using tools such as Adobe Photoshop and MS Paint. Modifications included text manipulation, image substitution, seal/stamp removal, and font or alignment inconsistencies. To emulate practical conditions, variations in lighting, background, and noise were also incorporated.

The dataset was partitioned into training (70%), validation (15%), and testing (15%) subsets:

Training Set: 1,050 images
Validation Set: 225 images
Testing Set: 225 images

All samples were stored in JPEG/PNG format at 300 DPI resolution. Annotations (genuine or forged) were manually verified to ensure labeling accuracy.

B. System Configuration

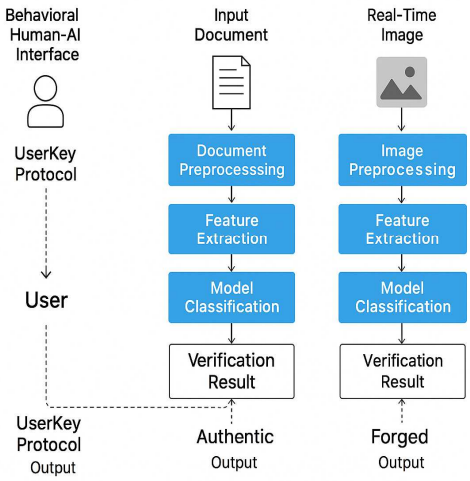
Development and evaluation were conducted on a workstation with the following specifications:

OS: Windows 11 Pro (64-bit)
CPU: Intel Core i7 (11th Gen) @ 2.80 GHz
RAM: 16 GB DDR4
GPU: NVIDIA GeForce GTX 1660Ti (6 GB)
Tools: Jupyter Notebook, Visual Studio Code
Libraries: Python, OpenCV, TensorFlow, Keras, Tesseract OCR, Scikit-learn

This configuration enabled efficient training and inference with GPU acceleration support.

C. System Architecture

Figure1. system structured pipeline:



III. SYSTEM OVERVIEW

A. Behavioral Human–AI Authentication Layer

The proposed AI-Enhanced Document Verification System is designed as a two-layered architecture consisting of the Behavioral Human-AI Interface (UserKey Protocol) and the AI-powered Document Verification Pipeline. These layers collectively ensure that verification is initiated only by authenticated users and that documents undergo rigorous automated analysis for authenticity.In the behavioral authentication stage, the UserKey Protocol validates the identity of the user before document analysis begins. This mechanism employs dynamically generated contextual questions, which are derived from personal preferences, past interactions, or real-time contextual factors such as device, location, or time of access. Examples include questions like *"Which of these would you never eat for breakfast?"* or *"What was the color of your last uploaded document?"*. These prompts may be single-choice or multi-choice, text-based or visual, and are often time-sensitive to identify hesitation or automated (bot-like) behavior. Responses are matched against the user's behavioral profile, and failure results in immediate termination of the verification process. Only authenticated users proceed to the document verification pipeline.The first layer ensures that only legitimate users initiate the verification process. A UserKey Protocol generates contextual questions based on user history, environment, and behavioral traits. Responses are matched against a stored profile using similarity functions.

Algorithm: Rule-based + Threshold Matching
Python Flask/Django (backend authentication)
SQLite/PostgreSQL (user profile storage)
Scikit-learn (similarity scoring, cosine similarity for textual answers)
Mathematical Model:
$A(u)=\{1,0,\text{if } f(Q,Ru)\geq\theta \text{ otherwise}$

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where Q represents contextual questions, Ru the user response, and θ the decision threshold.

B. DigiLocker Integration Layer

The system integrates with DigiLocker for secure retrieval of government-issued documents. Authentication is performed via OAuth 2.0, followed by access-token exchange. Verified documents are fetched using DigiLocker APIs such as GetIssuedFiles or Download.

Algorithm: OAuth 2.0 Token Exchange + API Request Handling

API/Library:

DigiLocker Authorized Partner APIs

requests (Python HTTP client)

Postman / APISetu SDK for endpoint testing

JSON/XML parsers for metadata extraction

Mathematical Abstraction:

D=APIDL(token)

where D denotes the retrieved document and APIDL is the DigiLocker API service.

B. Document Preprocessing

Once the user is authenticated, documents are standardized to ensure consistent analysis. Noise removal using Gaussian and median filtering eliminates scanning artifacts and background interference, thereby improving clarity [3]. Next, binarization converts the document into a black-and-white format, simplifying edge, font, and shape detection [1]. To maintain compatibility with CNN-based models, standardization resizes document images to a fixed resolution (e.g., 224 × 224 pixels) [2]. Finally, Optical Character Recognition (OCR) is applied to extract textual information, enabling detailed analysis of font style, alignment, and detection of textual tampering [8].

C. Feature Extraction

Following preprocessing, the system extracts multi-domain features to capture indicators of forgery. Textual features include font style, spacing, kerning, and alignment irregularities [4]. Visual features, such as signatures, seals, stamps, and watermarks, are cross-referenced with authentic reference templates to confirm their validity. Additionally, structural features capture layout-based attributes, such as margins, header/footer alignment, and the placement of logos or photographs, which helps identify cloning or tampering. The extracted features are concatenated into a unified feature set, forming the input to the classification model.

Following preprocessing, the system extracts multi-domain features to capture indicators of forgery.

Image-Based Features: Canny edge detection, contour mapping, and compression artifact detection.

Text-Based Features: OCR extraction (with Tesseract OCR) to analyze font style, alignment, spacing, kerning, and metadata irregularities [8].

Structural Features: Capture layout-based attributes, such as margins, block alignment, and header/footer placement [4].

Biometric Features: Face verification using FaceNet / ArcFace embeddings and signature/seal detection using SIFT/SURF descriptors.

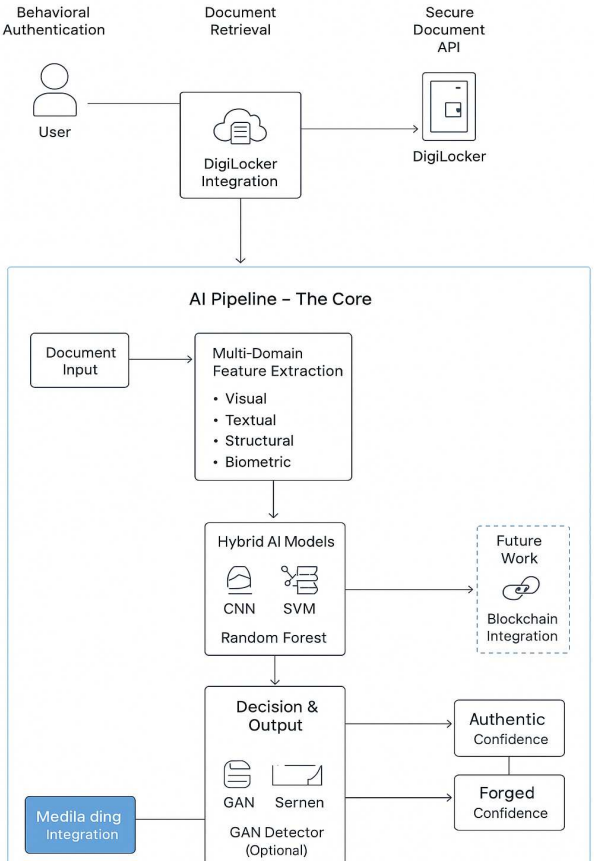
D. Model classification

The classification model is powered by a deep Convolutional Neural Network (CNN) trained offline on a labeled dataset containing both authentic and forged documents. The CNN architecture comprises three convolutional blocks with batch normalization and max pooling, followed by fully connected dense layers and dropout for regularization. Training is conducted using the Adam optimizer with a binary cross-entropy loss function, a batch size of 32, and up to 50 epochs with early stopping to prevent overfitting. To enhance generalization and robustness against subtle forgeries, the training pipeline incorporates data augmentation techniques such as random rotations, flipping, cropping, and brightness/contrast adjustments. This approach ensures resilience against varied forgery attempts, including minor manipulations.

E. Forgery Detection

During inference, the trained CNN processes the extracted features and produces classification results. Each document is labeled as either Authentic or Forged, accompanied by a confidence score that quantifies the model's certainty. Optionally, the system highlights specific regions suspected of forgery, such as tampered signatures, mismatched fonts, or irregular alignments, thereby enhancing interpretability and aiding human evaluators in decision-making.

Figure.2 AI-Powered Document Verification Process



i. Document Preprocessing

Once the document is retrieved, it is standardized to ensure consistent analysis.

Steps:

Normalization: Standardize document size and format.

Noise Removal: Use Gaussian and median filtering to eliminate scanning artifacts and background interference [3].

Deskewing: Correct document orientation using techniques like Hough Line Transformation.

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Resizing: Standardize to 224x224 pixels to maintain compatibility with CNN-based models [2].
Binarization: Convert the document into a black-and-white format to simplify edge, font, and shape detection [1].
Mathematical Model:
$D' = \phi(D)$ where $\phi(\cdot)$ represents preprocessing transformations applied on the original document D.
Mathematical Model: **$F = [Fi, Ft, Fs, Fb]$**
where Fi, Ft, Fs, and Fb represent image, text, structural, and biometric features respectively.

E. Forgery Detection and Classification

The extracted features are classified into "authentic" or "forged" using hybrid models.
Deep Learning Models:
CNN (primary classifier)
GAN Detector (deepfake detection)
Machine Learning Models:
SVM (text features)
Random Forest (metadata & structural features)
Training Details:
Optimizer: Adam ($\eta = 10^{-4}$)
Loss: Binary Cross-Entropy
Epochs: 50 with early stopping
Data Augmentation: rotation, flipping, brightness adjustment
Mathematical Loss Function:

$$L = -N1i = 1\sum N[yilog(y^i) + (1 - yi)log(1 - y^i)]$$

Decision Rule:

$$O(D) = \text{Authentic, Forged}, y \geq \tau y < \tau$$

where τ is the confidence threshold.

IV. EVALUATION

To evaluate the proposed Document Verification System, a dataset comprising 1,500 images of official documents was assembled. These documents were obtained from some publicly available sources, some were manually collected and some were synthetically generated to reflect realistic forgery scenarios. The dataset includes Aadhaarcards, PAN cards, driver's licenses, and academic certificates.

A. Data and System Configuration

Dataset:
1,500 images of official documents, including synthetically generated forgeries.
Partitioning: Training (70%), Validation (15%), and Testing (15%).
Annotations: Manually verified as either "genuine" or "forged."
System Configuration:
OS: Windows 11 Pro (64-bit)
CPU: Intel Core i7 (11th Gen) @ 2.80 GHz
RAM: 16 GB DDR4
GPU: NVIDIA GeForce GTX 1660Ti (6 GB)
Tools: Jupyter Notebook, Visual Studio Code
Libraries: Python, OpenCV, TensorFlow, Keras, Tesseract OCR, Scikit-learn

B. System Architecture and Results

The system follows a structured pipeline as shown below.
The proposed system achieves a classification accuracy of 98%, significantly outperforming traditional methods with lower processing time and better scalability, making it suitable for real-time enterprise applications.

System	Accuracy	Time (s)	Scalability
Proposed CNN	98%	0.3	HIGH
Baseline (Manual/OCR)	85%	2.0	Low

IV. Application Scenarios

The system is designed to integrate easily with enterprise workflows through RESTful APIs or lightweight mobile apps. It can be deployed in multiple real-world scenarios:
Banking KYC: Automating the validation of customer-submitted IDs during onboarding, reducing manual workload. Academic Institutions: Verifying certificates and transcripts submitted during admission processes to prevent document fraud.Legal Sector: Authenticating legal contracts and affidavits before filing or digital signing.Insurance and Claims: Detecting tampered medical bills, ID proofs, and receipts in fraud-prone insurance claims.
DigiLocker Integration (Top Layer): The unified DigiLocker and AI-powered document verification process integrates secure government-issued document retrieval with advanced artificial intelligence techniques for forgery detection. The process begins with the DigiLocker integration layer, where users authenticate through an OAuth 2.0 flow, and an access token is generated to securely fetch verified documents using DigiLocker APIs such as *GetIssuedFiles* or *Download*.
Document Input & Preprocessing: Once the document is retrieved in PDF, JPEG, or PNG format, it enters the AI pipeline for verification. The preprocessing stage ensures consistency by applying normalization, noise removal through Gaussian and median filtering, resizing the input to fixed dimensions (224 × 224 pixels), and converting the document into a binary format. Optical Character Recognition (OCR) is then employed to extract textual information, enabling analysis of fonts, alignment, and potential tampering.
Multi-Domain Feature Extraction: Following preprocessing, multi-domain feature extraction is performed to capture different indicators of authenticity. Visual features such as layout, fonts, watermarks, logos, and stamps are examined for irregularities, while photo analysis verifies the identity of the individual in ID documents using deep learning–based facial recognition models like FaceNet or ArcFace. Signatures and seals are checked against reference templates, and textual features are validated for font consistency, spacing, and regulatory formats. Structural features such as document margins, block alignment, and header placement are also analyzed to identify cloned or manipulated content.
AI Model Classification: The extracted features are then passed into a classification stage powered by deep learning and machine learning algorithms. Convolutional Neural Networks (CNNs) form the backbone of the system, detecting visual and structural anomalies, while Support Vector Machines (SVMs) and Random Forest classifiers enhance classification accuracy on textual and structured data. A GAN-detector module is also included to identify AI-generated or deepfake document forgeries. These models are trained using the Adam optimizer, binary cross-entropy loss function, and data augmentation techniques such as rotation, flipping, and brightness adjustments to improve robustness.

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Decision & Output: The final stage involves decision-making, where the system classifies the document as authentic or forged and provides a confidence score (e.g., 92.7%). A detailed report is generated, optionally highlighting suspected forged regions such as altered signatures, seals, or text. The output is presented to the user via an intuitive interface. To ensure continuous improvement, a feedback loop is incorporated, where verification results are stored and used to retrain the models, enhancing accuracy over time. Optionally, results can be logged on a blockchain to provide tamper-proof auditability.

This end-to-end workflow ensures that documents obtained from a trusted source like DigiLocker undergo rigorous AI-driven analysis, combining secure retrieval with deep learning–based forgery detection, making the system suitable for high-stakes applications in government, finance, and enterprise environments.

The proposed system achieves higher accuracy with lower processing time and better scalability, making it suitable for real-time enterprise applications.

Error Analysis

Minor false positives were observed, mainly in documents with unusual fonts or partially damaged regions. These can be mitigated through further model tuning or integration with attention-based architectures.

CONCLUSION AND FUTURE WORK

This paper presents an AI-powered document verification system capable of detecting forged documents with high accuracy and efficiency. By leveraging Convolutional Neural Networks (CNN) and advanced image processing techniques, system automatically identifies inconsistencies in fonts, natures, and embedded security features. Achieving a clas-cation accuracy of 98%, the proposed model significantly outperforms traditional methods and offers real-time, scalable verification for high-stakes applications. The modular design allows easy integration into enterprise systems and adaptability to various document types, including IDs, certificates, and legal records. Comparative evaluation with baseline systems confirms superior performance in terms of speed, precision, and scalability. Future work will focus on enhancing model robustness against deepfake and GAN-based forgeries by integrating attention mechanisms and transformer-based architectures. Blockchain integration is also planned to ensure tamper-proof and transparent document storage. Moreover, expanding the system to support mobile platforms, multilingual documents, and IoT-enabled edge devices will broaden its applicability in real-world scenario

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